

TM Forum Technical Specification

Product Usage Management

TMFC040

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Notice

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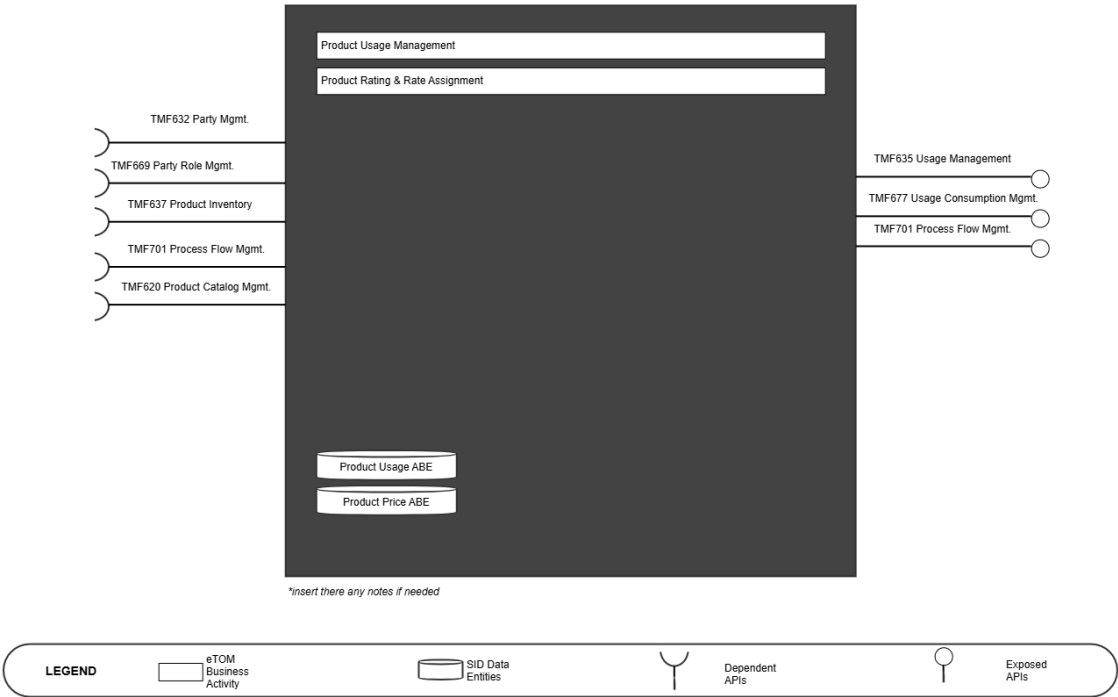
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1. Overview

Component Name	ID	Description	ODA Function Block
Product Usage Management	TMFC040	The Product Usage Components provides standardized mechanisms for product usage management (creation, update, retrieval, import and export of a collection of usages) and Product Rating & Rate Assignment by assigning a value (monetary or other) to an event in the context of a product, a party (customers and partners) and payer.	Core Commerce Management

TMFC040: Product Usage Management



2. eTOM Processes and SID Data Entities

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.2.16	L2	Product Usage Management	The Product Usage management processes encompass the functions required to guide, distribute, mediate, summarize, accumulate, and analyze Product Usage records. These processes may occur in real-time, near real-time (i.e. just at the end of the usage), or may be executed on a periodic basis. Based on Service Usage, this process aims at identifying Product Usage. For example, for a Video on Demand where you can watch a video as many time as you want during 72 hours, several Service Usages might have been tracked (each time the user watches the video) and only one Product Usage will be identified for all Service Usages in the 72 hours after the first watch. The guiding processes ensures that the Product Usage records used in the billing processes are appropriately related to the correct customer billing account and products. The Product Usage records are edited and if necessary reformatted (mediated) to meet the needs of subsequent processes. The billing event records may also be enriched with additional data during this process.
1.2.16.1	L3	Product Usages	The Enrich Product Usages processes will augment the product usage records by adding data to the records from sources such as customer, product, or other reference data.
1.2.16.1.1	L4	Add Product Usage Data	Add data to the records from sources such as customer, product, or other reference data to augment the product usage records.
1.2.16.1.2	L4	Assign Product Usage Price	Assign a price to product usage without consideration of specific product or customer information. The assigned price

Identifier	Level	Business Activity Name	Description
			may be used to enrich the product usage record.
1.2.16.2	L3	Guide and Assign Product Usages	The Guide Product Usages processes ensure that the event records used in the billing process relate to the correct customer billing account and products. A specific product usage record may be related to multiple customer billing accounts and subscribed products. Distribution of product usage records to other processes may also occur.
1.2.16.2.1	L4	Assign Product Usages	Ensure that the Product Usages used in the billing process relate to the correct Product.
1.2.16.2.2	L4	Distribute Product Usage	Distribute billing event records to other processes.
1.2.16.2.3	L4	Guide Product Usages	Guide Product Usages process is in charge of identifying Product Usages based on Service Usages. For example, for a Video on Demand where you can watch a video as many time as you want during 72 hours, several Service Usages might have been tracked (each time the user watches the video) and only one Product Usage will be identified for all Service Usages in the 72 hours after the first watch.
1.2.16.3	L3	Mediate Product Usages	The Mediate Product Usages process edits and reformats the data record to meet the needs of a recipient application.
1.2.16.3.1	L4	Edit Product Usages	Edit the data record for recipient applications.
1.2.16.3.2	L4	Reformat Product Usages	Reformat the data record for recipient applications.
1.2.16.4	L3	Report Product Usage Records	The purpose of the Report Product Usage Record processes is to generate reports on Product Usage records based on requests from other processes. These processes produce reports that may identify abnormalities, which may be caused by fraudulent activity or related to customer complaints. Investigation of problems related to these product usage records is also part of this process. These processes also support other processes such as customer review of

Identifier	Level	Business Activity Name	Description
			product usages (pre-billing and post-billing).
1.2.16.4.1	L4	Generate Product Usage Report	Generate reports on product usage records based on requests from other processes.
1.2.16.4.2	L4	Investigate Product Usage Related Problem	Investigate problems related to product usage records.
1.2.16.4.3	L4	Support Product Usage Related Process	Support other processes such as customer review of product usages (pre-billing and post-billing).
1.2.17	L2	Product Rating & Rate Assignment	The purpose of Product Rating & Assignment is to rate a value (monetary or other) to Product Usage or a set of Product Usages and assign the result to a Product and a Billing Account. The charge may be either a credit or a debit and can be handled either online or offline. Online charging is performed in real-time, requiring an authorization component which may affect how the service is rendered and enables an operator to provide prepaid services to its customers. Whereas offline charging is performed after the service is rendered and is not required to be done in real-time and generally relates to subscription based products.
1.2.17.1	L3	Perform Rating	Process responsible for calculating the value of a product usage or a set of product usages, before, during or after the rendering of the service, based on parameters of the request (type, quantity, etc.), parameters of the customer/subscriber (tariffs, price plans, accumulated usage, contracts, etc.) and other parameters (time-of-day, taxes, etc.). The same request maybe rated differently for different subscribers based on their purchased offers or agreements.
1.2.17.2	L3	Aggregate Items For Rate Assignment	This process is responsible for accumulating contributing items, which can be quantities, values (monetary or other) or both. Aggregation can occur over time or can be initiated to gather a “snapshot” of the items at a point in time.
1.2.17.3	L3	Manage Customer Assignment Hierarchy Managing the charging relationships among subscribers.	Customer hierarchies are commonly used for corporate customers, family plans or other type of affinity groups. This process manages the assignment relationships among subscribers, e.g. sharing, inheriting or restricting balances, price plans and

Identifier	Level	Business Activity Name	Description
			discounts. Thereby assuring that a charge is added to or subtracted from the correct account balance.
1.2.17.4	L3	Provide Advice of Rate	The activity of Provide Advice of Rate (aka Advice of Charge) is responsible for providing advice on rates, in real-time or offline, an estimate or value of the rate for a specific usage request. The advice is usually based upon performing a full rating process for the request.
1.2.17.5	L3	Apply Rate Level Discounts	This process applies discounts to product prices at an individual product level. A discount may be expressed as a monetary amount or percentage, and modifies a price for a product. When a discount is expressed as a percentage, the discounting process determines the discount calculated in relation to the price for the product. The discount may be displayed as a separate entry on the bill or may be combined with the rate for the product to only show as one entry. Discounts may be a one-time event or may have some duration (days, months, life of product, etc.). Discounts may apply to a specific customer or be generally available based on selection of products (for example - bundles). Discounting structures may involve tiers, tapers, or thresholds.

2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Product Usage ABE**	
Product Price ABE***	

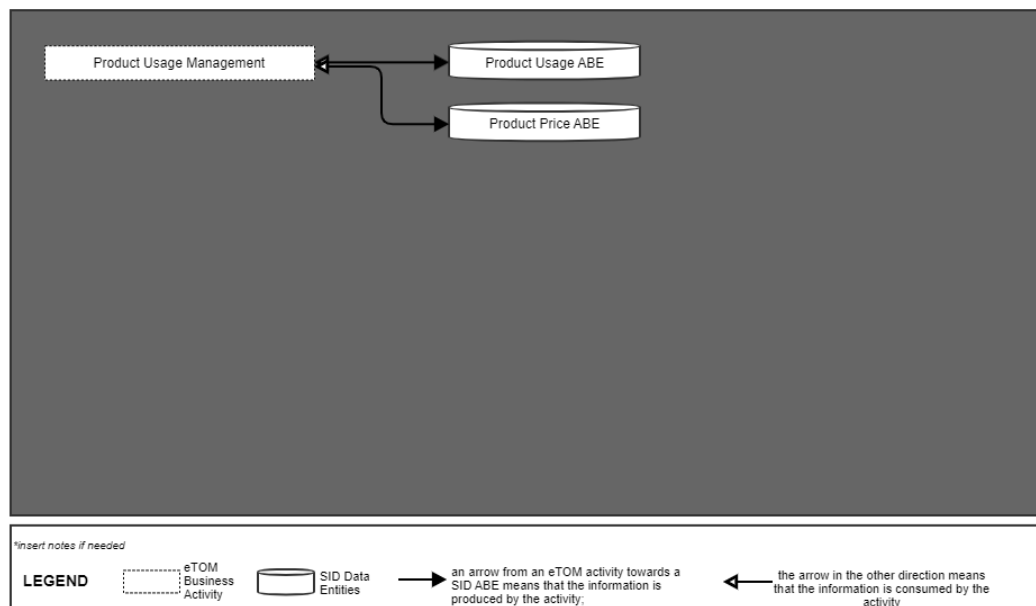
*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

** In the context of this component, it relates to the usage events and not the management of the ProductUsageSpecification. The ProductUsageSpecification is maintained by TMFC001 Product Catalog Management component.

*** In the context of Usage; related to UsageProdPriceCharge and related ProdPriceAlteration.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



3. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
253	Customer Usage and Charges Report Access	Customer Usage and Charges Report Access provides an internet technology driven interface to the customer to access web based reports for (historical) usage and charges directly for themselves.	Product Usage Management Rating and Follow up	Rating and Follow up Bill Calculation
55	Price and Discount Calculation	Price and Discount Calculation applies pricing and discounting rules and algorithms in the context of the assembled information concerning Products (i.e. instances of Product).	Product Usage Management Rating and Follow up	Rating and Follow up Tariff Calculation and Rating
125	Charging Event Accumulation	Charging Event Accumulation function accumulates events that provide measurements that will be used in the charge calculation (e.g. used allowance).	Product Usage Management Rating and Follow up	Rating and Follow up Tariff Calculation and Rating
126	Event Charge/Credit Calculation	Event Charge/Credit Calculation calculates event-level charges/credits (one time, recurring, and usage).	Product Usage Management Rating and Follow up	Rating and Follow up Tariff Calculation and Rating
127	Calculated Charges/Credits Proration	Calculated Charges/Credits Proration provides proration of calculated charges/credits. The function handles partial rating of a period.	Product Usage Management Rating and Follow up	Rating and Follow up Tariff Calculation and Rating
128	Late Arrival Usage Charges/Credits Recalculation	Late Arrival Usage Charges/Credits Recalculation provides recalculation of	Product Usage Management Rating and Follow up	Rating and Follow up Tariff Calculation and Rating

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		charges/credits based on information received later (e.g. from the Service Level Agreement function, delayed call detail record file arrival, delayed order arrival). Recalculation may be necessary: pre-billing (prior to Bill Calculation), during the Bill Calculation process, and/or post-billing.		
186	Charging/Rating Recalculation	Charging/Rating Recalculation function recalculates the charges, when appropriate, across product, location, or customer, and considerations based on business rules.	Product Usage Management Rating and Follow up	Rating and Follow-up Tariff Calculation and Rating
300	Discounts Calculation	Discounts Calculation determines charge discounts based on pricing plan; including discounts on recurring, one time, and usage charges. Discounts may be applied at different levels such as cross product, cross location, or cross customer (all customers that are part of a given group plan – some affiliation). The discounts can be apportioned across multiple events.	Product Usage Management Rating and Follow up	Rating and Follow-up Tariff Calculation and Rating
67	Usage Summary and Details Presentation	Usage Summary and Details Presentation presents usage summary and details (billed and non-billed) for a specific time period.**	Invoice Management	Invoicing

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
85	Billing Event Processing Distribution	Billing Event Processing Distribution function for correlation and distribution of usage data for processing of e.g. bill, customer and product. To supply relevant usage data to the Billing System.	Invoice Management	Invoicing
86	Billing Event Processing Enrichment	Billing Event Processing Enrichment with complementing data e.g. product or location information to supply relevant information to the Billing System.	Invoice Management	Invoicing

** Consumption is exposed through TMF677 and Usage details through TMF635.

4. TMF OPEN APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

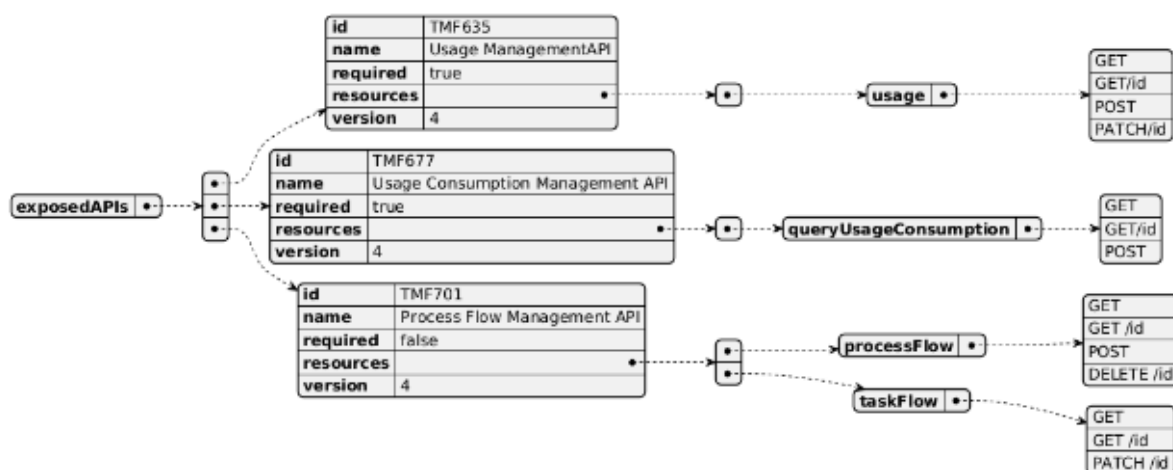
The Product Usage Management Component relate to usage and the API TMF635 Usage Management is in evolution as it mix concerns on product, service and resource levels. With version 5 of Open APIs, there is a major evolution of Usage APIs so that there will be a distinct usage API per Product, Service, and Resource:

- TMF767 Product Usage
- TMF727 Service Usage
- TMF771 Resource Usage

These will replace the existing TMF635 Usage Management that will stop with v4 and not be migrated to v5. The Product Usage Management component relates to the usage on Product Level so in future iterations, the TMF767 will be included.

4.1. Exposed APIs

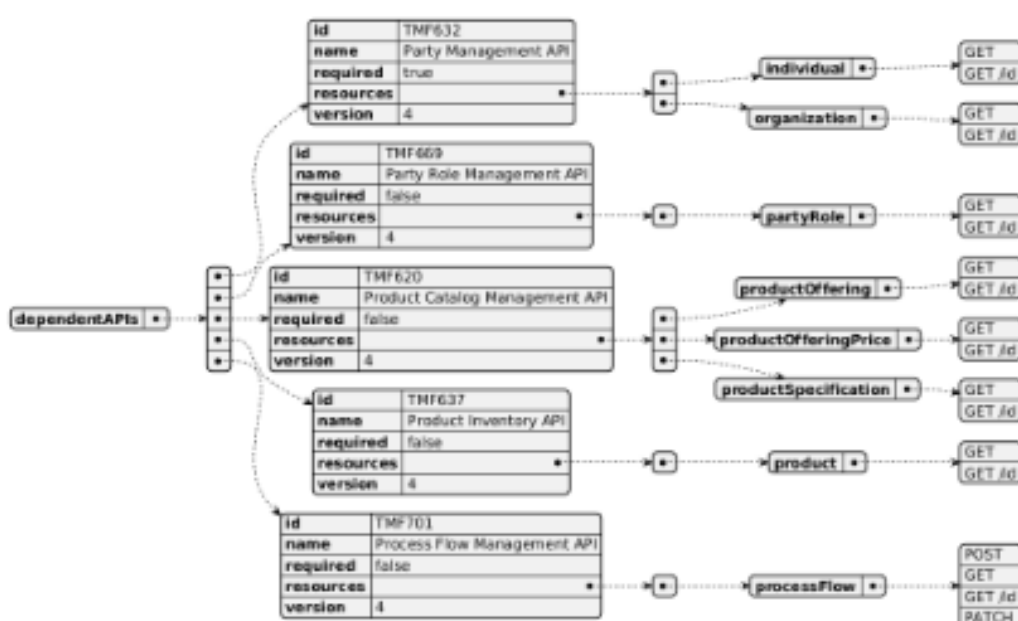
Following diagram illustrates API/Resource/Operation:



API ID	API Name	Mandatory / Optional	Operations
TMF635	Usage Management	Mandatory	usage: GET, GET /id, POST, PATCH/id
TMF677	Usage Consumption Management	Mandatory	queryUsageConsumption: GET, GET /id, POST
TMF701	Process Flow Management API	Optional	processFlow: GET, GET /id, POST, DELETE /id taskFlow: GET, GET /id, PATCH /id

4.2. Dependant APIs

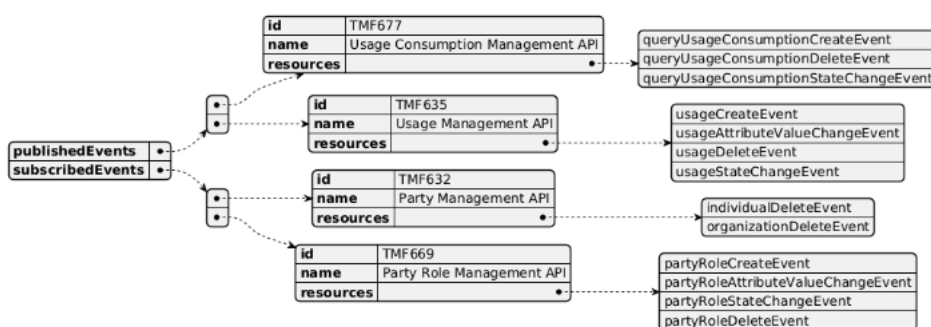
Following diagram illustrates API/Resource/Operation potentially used by the product catalog component:



API ID	API Name	Mandatory / Optional	Operations	Rationale
TMF632	Party Management	Mandatory	individual: GET, GET /id organization: GET, GET /id	To be able to guide the Product Usage to the appropriate Party and its related products.
TMF669	Party Role Management	Optional	partyRole: GET, GET /id	
TMF620	Product Catalog Management API	Mandatory Optional	productOffering: GET, GET /id productOfferingPrice: GET, GET /id productSpecification: GET, GET /id	
TMF637	Product Inventory	Mandatory	product: GET, GET /id	To retrieve installed product and price information to determine the rate to assign.
TMF701	Process Flow Management	Optional	processFlow: GET, GET /id, POST, PATCH	

4.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



5. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

6. References

6.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

6.2. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

7. Administrative Appendix

7.1. Document History

7.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	13-Feb-2024	Gaetano Biancardi Elisabeth Andersson	Final edits prior to publication
1.1.0	08-Oct-2024	Elisabeth Andersson	Updated to the latest version of the Component Template. Added comment and Jira tickets related to Usage API evolution of v5. Clarified the scope of SID for this component in relation to ProductUsageSpecification. Changed requirements for TMF620 and TMF672 from Mandatory to Optional. Mapped to the latest Frameworks with the follow changes: Functional Framework: - Name change of Aggregation level 1 and 2: 253, 55, 125, 126, 127, 128, 186, 300 - Added functions: 65, 85, 86 Business Process Framework: - Business Process Name change: 1.2.17.3
1.1.0	14 Nov 2024	Amaia White	Final edits prior to publication

7.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	13-Feb-2024	Gaetano Biancardi Elisabeth Andersson	Initial release of document
Production	26-Apr-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status.

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	14 Nov 2024	Amaia White	Release of version 1.1.0
Production	27-Dec-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

7.2. Acknowledgments

Team Member (@mention)	Title	Company	Role*
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